

NMap (Network Mapper)
tool used by hackers to scan network

first step

★ ping - networking command used to check whether your computer can reach another device or website over a network

ping 8.8.8.8
ping google.com

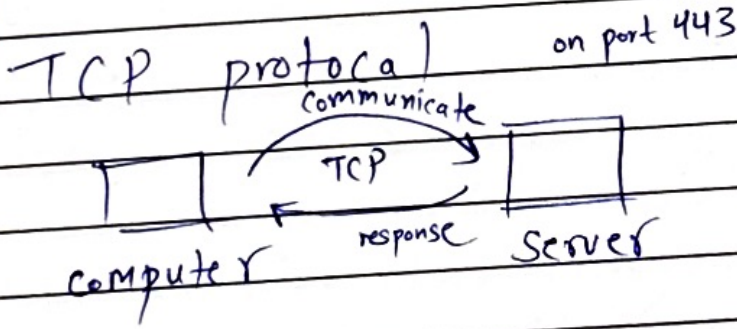
Second Step

look for ports 80, 443

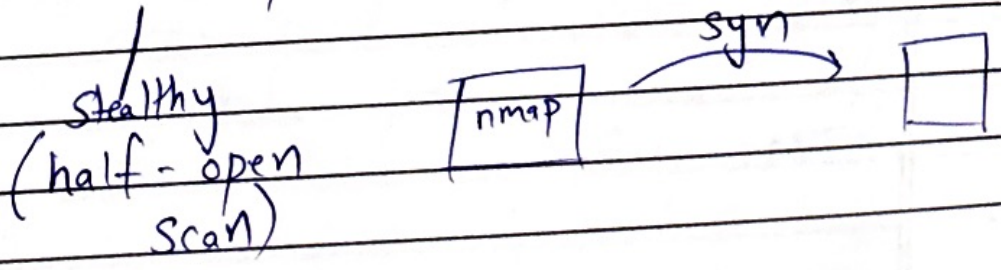
sudo nmap -sT -p 80,443 192.168.29.57/24

TCP Connect
(full open scan)
3 way handshake

CVE - Common vulnerabilities and exposures



sudo nmap -sS -p 80,443 192.168.29.57



steps

OS
detection

`sudo nmap -O 192.168.1.10`

step 4

Aggressive scan — includes OS detection, version detection, traceroute and scripts.

steps

Save results

`nmap -sV 192.168.1.10 -oN scan.txt`

Demo —

1. Start both Kali & Metasploitable
2. Find metasploitable's IP
if config
3. From Kali run `ping <Metasploitable IP>`
4. If ping works
`nmap -sV <metasploitable -IP>`
5. Show open ports & services found.

```
(navya@kali)-[~]
```

```
$ man nmap
```

```
(navya@kali)-[~]
```

```
$
```

```
(navya@kali)-[~]
```

```
$ sudo nmap -A 192.168.29.57
```

Starting Nmap 7.98 (<https://nmap.org>) at 2026-06-07 13:12 +0530

Nmap scan report for 192.168.29.57

Host is up (0.0018s latency).

Not shown: 997 filtered tcp ports (no-response)

PORT	STATE	SERVICE	VERSION
135/tcp	open	msrpc	Microsoft Windows RPC
139/tcp	open	netbios-ssn	Microsoft Windows netbios-ssn
445/tcp	open	microsoft-ds?	

Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port

Device type: VoIP adapter|bridge|general purpose

Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)

OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu

Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode n

No exact OS matches for host (test conditions non-ideal).

Network Distance: 1 hop

Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:

```
| smb2-time:
|   date: 2026-06-07T07:42:21
|_  start_date: N/A
|_ clock-skew: -2s
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled and required
```

TRACEROUTE (using port 139/tcp)

HOP	RTT	ADDRESS
1	5.09 ms	192.168.29.57

OS and Service detection performed. Please report any incorrect results at <https://nmap.org/submit/>.

Nmap done: 1 IP address (1 host up) scanned in 24.86 seconds

```
(navya@kali)-[~]
```

```
$
```

```
30 00:00:05.313766 file sy
31 00:00:05.313795 File sy
32 00:00:05.358185 Shared
```